INCH-POUND

MIL-DTL-19246E
05 February 2007
SUPERSEDING
MIL-DTL-19246D
27 December 2000

DETAIL SPECIFICATION

CONNECTOR, OXYGEN MASK HOSE, TYPE MC-3A

Reactivated after 05 February 2007 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers type MC-3A oxygen mask hose connector for connecting the oxygen mask hose to the oxygen regulator hose.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS22016	-	Connector, Oxygen Mask Hose, Type MC-3A. (Inactive for New Design)
MS22058	-	Connector, Oxygen Hose to Regulator
MIL-PRF-27210	-	Oxygen, Aviator's Breathing, Liquid and Gas

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals

(Copies of these documents and standards are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4	-	Procedures, Sampling and Tables for Inspection by Attributes. (DoD adopted)

(Copies of the above document are available from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53201-4606 or http://www.asq.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D1149	-	Rubber Deterioration – Surface Ozone Cracking in a Chamber. (DoD adopted)

(Copies of the above document are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or http://www.astm.org.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The connector furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and in MS22016 and MS22058.

3.3.1 Nonmetallic materials. Nonmetallic materials, except for the poppet material as shown on MS22016, shall be of an ozone-resistant composition.

3.3.1.1 Age. Age of elastomer components shall be not greater than 12 months from the date of manufacture to the date of delivery to any Government service or to any airframe or accessory manufacturer.

3.3.2 Metal Parts. All metal parts shall be of a corrosion resistant material or treated to render them resistant to corrosion.

3.3.2.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals, in accordance with MIL-STD-889, shall not be in intimate contact with each other.

3.3.3 Protective treatment. When materials used in the construction of the connector assembly are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration that shall in no way prevent compliance with the performance requirements (see 3.5) of this specification. Protective coatings which crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.4 Design and construction. The design and construction of the connector assembly shall be in accordance with MS22016.

3.4.1 Connector assembly valves. Connector assembly valves shall make it possible for the user to breathe normally through the connector when the valve is inserted into a connector conforming to MS22058. When disconnected, the valve shall permit inhalation, but a noticeable

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resistance shall be introduced to indicate that disconnection has occurred. A spring-loaded relief valve shall be located in the connector body and adjusted to allow oxygen from the bailout cylinder to escape whenever the mask pressure exceeds a pressure equal to 12 inches of water greater than ambient pressure. The relief valve shall vent small flows before the mask pressure reaches 12 inches of water greater than ambient pressure.

3.4.2 Bailout attachment. The side bailout attachment shall contain a check valve as shown on MS22016.

3.5 Performance.

3.5.1 Disconnection force. The connector shall disconnect between 12 and 20 pounds when tested as specified in 4.7.2.

3.5.2 Leakage (room temperature and extreme temperature). Leakage of the connector assembly shall be not greater than 0.01 liter per minute (Lpm) when tested as specified in 4.7.3, 4.7.9 and 4.7.10.

3.5.3 Relief valve setting. The connector shall meet the following requirements when tested as specified in 4.7.4:

Flow (Lpm)	Developed Pressure (inches of water)
2.5	less than 12
12.0	between 11 and 15
50.0	between 12 and 18

3.5.4 Pressure drop through MC-3A connector. The pressure drop through the connector shall be not less than 4 inches nor greater than 6 inches of water when tested as specified in 4.7.5.

3.5.5 Pressure drop through MC-3A connector with MS22058 connector. The pressure drop through the connector assembly shall be not greater than the following values when tested as specified in 4.7.6:

Flow (Lpm)	Pressure Drop (inches of water)
135	1.50
90	0.66
70	0.40
50	0.20
30	0.07

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3.5.6 Proof pressure. The connector shall not show any evidence of structural damage when tested as specified in 4.7.7.

3.5.7 Bailout flow. The connector shall permit a minimum flow of 9 Lpm when tested as specified in 4.7.8.

3.5.8 Endurance. The connector shall disconnect between 12 and 20 pounds when tested as specified in 4.7.11.

3.5.9 Drop test. The connector shall operate when tested as specified in 4.7.12.

3.5.10 Salt fog. The connector shall not show any evidence of corrosion when tested as specified in 4.7.13.

3.5.11 Disassembly and examination. The connector shall not show evidence of deterioration or wear when tested as specified in 4.7.14.

3.5.12 Ozone-resistance. The elastomer components shall not bloom or show any evidence of checking, cracking, or other damage when tested as specified in 4.7.15.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Weight. The weight of the complete connector shall be not greater than 0.20 pound.

3.8 Identification of product. The connectors shall be marked for identification in accordance with MIL-STD-130.

3.9 Marking. Nomenclature tags shall be inserted in each bag showing title, part number or other data as required by the procuring activity. In addition, interior and exterior containers shall be marked in accordance with MIL-STD-129.

3.9.1 Precautionary marking. The following precautionary marking shall appear on each package:

CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE OXYGEN CONNECTORS.

3.10 Workmanship. The connectors shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could affect safety, performance, reliability, or durability.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. First article inspection (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Disconnection force
Leakage (room temperature)
Relief valve setting
Pressure drop through MC-3A connector
Pressure drop through MC-3A connector with MS22058 connector
Proof pressure
Bailout flow
Drop test

4.3 First article inspection. First article inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Disconnection force
Leakage (room temperature)
Relief valve setting
Pressure drop through MC-3A connector
Pressure drop through MC-3A connector with MS22058 connector
Proof pressure
Bailout flow
Drop test

4.3.1 First article samples. After the award of the contract or order, the manufacturer shall submit three connector assemblies. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these connector assemblies from contract to contract, submission of

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further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.4.1). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirement of submitting to the conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.4.1).

4.4 Conformance inspection. Conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Disconnection force
Leakage (room temperature)
Relief valve setting
Pressure drop through MC-3A connector
Pressure drop through MC-3A connector with MS22058 connector
Bailout flow

4.5 Inspection lot. An inspection lot size shall be expressed in units of one connector made under essentially the same conditions and from the same materials and components. The sample unit shall be one connector.

4.5.1 Sampling for tests and examinations of connector assemblies. The sample size, acceptance criteria, tests, and examinations required for the connector assemblies shall be as specified in table I.

TABLE I. Sample size, test and examinations.

INSPECTION	TYPE OF INSPECTION	TEST METHOD	SAMPLE SIZE 1/
Visual Examination (See table II, Classification of defects)	Critical	4.7	Every connector assembly
	Minor		Inspection Level II
Dimensions	Major	4.7.1	Inspection Level S-2
Disconnection force	Critical	4.7.2	Every connector assembly
Leakage (room temperature)	Critical	4.7.3	Every connector assembly
Relief valve setting	Critical	4.7.4	Every connector assembly
Pressure drop through connector	Critical	4.7.5	Every connector assembly
Pressure drop MC-3A connector with MS22058 connector	Major	4.7.6	Inspection, Level S-2
Bailout flow	Major	4.7.8	Inspection Level S-2

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of ASQ-Z1.4.

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4.6 Gas conditions. The gas used in testing the connector assemblies shall be oxygen conforming to MIL-PRF-27210, type I.

4.6.1 Temperature and pressure conditions. Unless otherwise specified, inspections shall be conducted at local ambient temperatures and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by any instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP requirements. NTP conditions shall be 29.92 inches of mercury and 70 °F.

4.7. Visual examination. Every connector shall be examined visually for critical defects to determine conformance to this specification and to MS22016. Table II shall be used to classify the defects found.

TABLE II. Classification of defects.

CRITICAL	MINOR
1. Material imperfections - foreign matter embedded. 2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws. 3. Any component missing, malformed, fractured, or otherwise damaged. 4. Any component loose or otherwise not securely retained. 5. Incorrect assembling or improper positioning of components. 6. Any functioning part that works with difficulty. 7. Faulty workmanship or other irregularities.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent.

4.7.1 Dimensions. The connector shall be checked dimensionally to determine conformance to MS22016.

4.7.2 Disconnection force. The connector shall be installed in a connector assembly conforming to MS22058. The connector assembly shall then be subjected to a straight pull parallel to the longitudinal axis. An average of five pulls conducted at five second intervals shall

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be used as the disconnection force. The disconnection force shall meet the requirements specified in 3.5.1.

4.7.3 Leakage (room temperature). The connector shall be installed in a connector assembly conforming to MS22058. An internal pressure of one psi at room temperature shall be applied to the connector assembly while it is subjected to a pull of not greater than 2.0 pounds applied in any direction. The leakage shall meet the requirements specified in 3.5.2.

4.7.4 Relief valve setting. To measure the developed pressure, a piezometer ring shall be located at 2 ± 0.5 inches from the oxygen mask end of the connector. Pressure shall be applied to the oxygen mask end of the connector to obtain flows of 2.5, 12, and 50 Lpm. The developed pressure shall then be measured at the specified flows and meet the requirements specified in 3.5.3.

4.7.5 Pressure drop through MC-3A connector. A piezometer ring shall be located at 2 ± 0.5 inches from the end of the connector. The ring shall be located on the mask end if the suction end is used or on the regulator end if pressure is used. The flow duct shall be constructed to ensure a straight flow for at least 6 inches immediately before entering the connector if pressure is used or immediately after leaving the connector if suction is used. The connector shall be mounted in a vertical position, mask end up, and located so as to permit a free flow of room air out of the connector if pressure is used or into the connector if suction is used. A pressure or suction shall be applied to the connector to obtain a flow of 15 Lpm from the regulator end to the mask end. The pressure drop through the connector shall then be measured by means of the piezometer ring and shall meet the requirements specified in 3.5.4.

4.7.6 Pressure drop through MC-3A connector with MS22058. A piezometer ring shall be located 2 ± 0.5 inches from the applicable end of the connector. For instructions concerning the use of the piezometer ring see 4.7.5. The MC-3A connector shall then be installed in a connector conforming to MS22058. Pressure or suction shall be applied to the connector to obtain flows of 30, 50, 70, 90, and 135 Lpm from the regulator end to the mask end. The pressure drop through the assembly shall then be measured by means of the piezometer ring and shall meet the requirements specified in 3.5.5.

4.7.7 Proof pressure. The connector shall be installed in a connector conforming to MS22058. The connector assembly shall be subjected to an internal pressure of 5 psi for a period of 3 minutes. The connector shall pass the performance requirement specified in 3.5.6. The connector shall then be subjected to and pass the tests specified in 4.7.4 and 4.7.6.

4.7.8 Bailout flow. Attach the hose of an 1800 psi H-2 emergency bailout bottle to the emergency oxygen port of a MC-3A connector and connect a flowmeter to the mask end of the connector. The H-2 emergency bailout bottle shall be activated and the flow of oxygen through the connector measured. The bailout flow through the connector shall meet the requirements specified in 3.5.7.

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4.7.9 Leakage (high temperature). The connector shall be conditioned at a temperature of 160 ± 2°F for a period of 2 hours. After conditioning, and while still at this temperature, the connector shall be subjected to and pass the leakage test specified in 4.7.3.

4.7.10 Leakage (low temperature). The connector shall be conditioned at a temperature of -65 ± 2 °F for a period of 2 hours. After conditioning, and while still at this temperature, the connector shall be subjected to and pass the leakage test specified in 4.7.3.

4.7.11 Endurance. The connector shall be inserted into and disconnected from a connector conforming to MS22058 for a total of 1,000 times each. The insertions and disconnections shall be conducted at the rate of one complete cycle every five seconds. After 1,000 cycles have been completed, the connector shall remain at rest for a period of 24 hours. The connector shall then be subjected to the test specified in 4.7.2 to determine the disconnection force after endurance. The disconnection force shall meet the requirements specified in 3.5.8. The connector shall then be subjected to and pass the tests specified in 4.7.3 and 4.7.4.

4.7.12 Drop test. The connector shall be subjected to a free fall of 2 feet so that the end that connects with the MS22058 connector hits onto concrete or an equally firm surface. This procedure shall be repeated for a total of 100 times. The connector shall then be subjected to and pass the tests specified in 4.7.3, 4.7.4, and 4.7.6.

4.7.13 Salt fog. Both ends of the connector shall be unplugged exposing the interior as well as the exterior of the connector. The connector shall be subjected to the salt fog test specified in MIL-STD-810, Method 509.3. The connector shall pass the performance requirement specified in 3.5.10. Upon completion of the salt fog test, the connector shall then be subjected to and pass the tests specified in 4.7.2, 4.7.3, 4.7.4, and 4.7.6.

4.7.14 Disassembly and examination. At the conclusion of all the above inspections the connector shall be disassembled and examined. The connector shall pass the performance requirements specified in 3.5.11. The connector shall be disassembled and examined after any test in which connector failure indicates faulty construction.

4.7.15 Ozone-resistance test (elastomer components). Three elastomer test slabs shall be tested for ozone-resistance. The test apparatus shall be in accordance with ASTM-D1149. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to 100 ± 2 °F and have an ozone concentration of 120 ± 10 parts by volume of ozone per million by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall be examined under 10x magnification and shall meet the performance requirements specified in 3.5.12.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Service's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The connector attaches the oxygen mask hose to the oxygen regulator hose. The oxygen mask and oxygen breathing equipment are unique to military aircraft because they must be able to withstand an emergency bailout pressure of 1,800 psi and still provide oxygen flow through the connector.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether first article inspection is required and where samples should be sent (see 3.2 and 6.4.1).

c. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19246 whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Air Systems Command, Code 4.6.2.2, 48110 Shaw Road, Bldg 2187, Patuxent River, MD 20670-5304.

6.4 First article. When first article is required, the connector will be tested under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a pre-production sample. The contracting officer should include specific

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instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

6.4.1 First article samples. First article samples should be sent to the Naval Air Systems Command, Code 4.6.2.2, Bldg 2187, 48110 Shaw Road, Patuxent River, MD 20670-5304.

6.4.2 Completion of the first article inspection. Applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production should be forwarded to the Government quality control representative. One approved connector will be returned to the manufacturer for use in monitoring production. The other connectors will be consumed or destroyed in the first article inspection and will not be considered as part of the quantity to be delivered under contract.

6.5 Subject term (key word) listing.

Breathing
High pressure
Low pressure
Regulator

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

(Project 1660-2006-003)

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